stat 5

variable, $X \longrightarrow x_1, x_2, \ldots, x_n$

(random variable) $Y \longrightarrow y_1, y_2, \ldots, y_n$

$\downarrow \quad \downarrow \quad \downarrow$

$p_1 \quad p_2, \ldots \quad p_n$

probability distribution

* Sample space (coin) : 2^{n} → No. of times

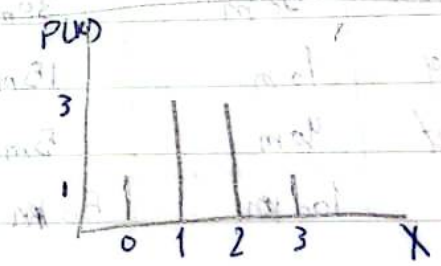
sample space (dice) : 6^{n}

لو رميت coin ٣ مرات وعايز ال head

Result	Coin toss 1	2	3	No of head
1	T	T	T	0
2	T	T	H	1
3	T	H	T	1
4	H	T	T	1
5	T	H	H	2
6	H	T	H	2
7	H	H	T	2
8	H	H	H	3

the probability dist of No. of heads

No of heads X	P(x)
0	$\frac{1}{8}$ = 0.125
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$
total	1



mean, S.D ← measures

① $0 \leq P(X) \leq 1$

② outcomes are mutually exclusive

③ $\sum P(x) = 1$

characteristics of distribution.

random variable
- discrete r.v (counting) (points)
- continous r.v (measuring) (Range)

Characteristics of discrete r.v:

① $0 \leq P(x) \leq 1$

② outcomes are mutually exclusive.

③ $\sum P(x) = 1$

④ $\mu = \sum X \cdot P(x)$ (mean)

⑤ $\sigma^2 = \sum (X-\mu)^2 \cdot P(X)$

~~*note what is the meaning of variance? the mean of the sum of the square of the distance betwee~~

assignment: find the mean and variance and standard deviation of the previous ex

$f(x) = P(X=x)$ → discreate (P.m.f) mass

→ Continous (P.d.f) density

→ Commulative distribution function:

$$F(x) = P(X \le x)$$

3 Characterestics:

① $F(x) = P(X \le x) = \sum_{X \le x} f(x)$

② $0 \le F(x) \le 1$

③ If $x \le y$, then $F(x) \le F(y)$

discrete distribution:

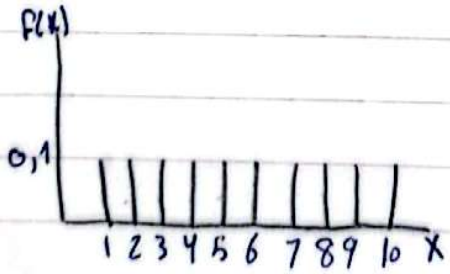
① uniform discrete dist.

if $X \sim$ uniform(n)

"is distributed"

$f(x) = \frac{1}{n}$ $\quad x = \{0, 1, 2, 3 \ldots, n\}$

$f(x) = 0,1$



$X \rightarrow a, 1+a, 2+a, \ldots, b$
$a \leq b$

① $\mu = E(X) = \frac{a + b}{2}$ (expected, min, max)

② $\sigma^2 = \frac{(b-a+1)^2 - 1}{12}$

③ $\sqrt{\frac{(b-a+1)^2 - 1}{12}}$

bionominal discrete distribution

- Bernoulli trial
  - success P
  - failure $1-P = q$

$P(H) = S$
$P(T) = F$

if $X \sim$ Bionominal (P, n)

3 characteristics:

① each trial has only 2 outputs (success or fail)
② $P(S)$ [probability of success] $P(S)$ is fixed from trial to trial
③ the trials is independent.

$\Rightarrow f(x) = {}_nC_x \, P^x q^{n-x}$

${}_nC_x \rightarrow$ combinations of n taken x at a time
$n \rightarrow$ number of trials
$x \rightarrow$ - - - - successes
$P \rightarrow$ the probability
$q \rightarrow$ number of failer $1-P$

① $\mu = np$

② $\sigma^2 = npq$

③ $\sigma = \sqrt{npq}$

$n = 3$

$P = \frac{1}{4} = 0{,}25$

$q = 0{,}75 = \frac{3}{4}$

ex

X	P(X)
0	$3C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^3 = \frac{27}{64}$
1	$3C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^2 = \frac{27}{64}$
2	$3C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right) = \frac{9}{64}$
3	$3C_3 \left(\frac{1}{4}\right)^3 = \frac{1}{64}$
total	1

① in the previous find mean, variance, σ

② propability of $P(X) \geq 2$ at least

atmost

③ $P(X) \leq 2$

* Geometric dist.

if $X \sim \text{Geometric}(P)$

$0 < P < 1$

$$f(x) = (1-P)^{x-1} P, \quad x = 1, 2, 3, \ldots$$

@ $P = 0{,}01 \qquad X = 125$

find $P(X = 125) = (0{,}99)^{124} \, 0{,}01 = 0{,}0029$

$E(X)$ ✓

$E(x)$ ✗

* $E(X) = \dfrac{1}{P}$

* $Var(X) = \dfrac{1-P}{P^2}$

* Poisson dist. « rare events »

$$f(x) = \frac{e^{-\lambda} \, \lambda^{x}}{x!}$$

- e is a natural number
- λ is the average of x
- $x!$ is the functional number of x

$x = 0, 1, 2, 3, \ldots$

@ $\lambda = 10 \qquad x = 12$

$$f(x) = \frac{e^{-10} \, 10^{12}}{12!} = 0{,}0095$$

Compute that $P(X \leq 20) = P(X=0) + P(X=1) + P(X=2) \cdots P(X=12)$

$$= \sum_{i=0}^{12} \frac{e^{-10} 10^{i}}{i!}$$

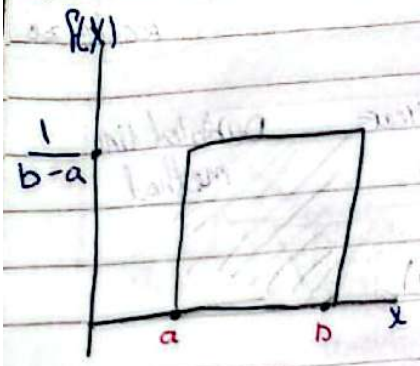
assignment

* $E(X) = \lambda$

$var(X) = \lambda$

$S.d(X) = \sqrt{\lambda}$

→ uniform continous dist.

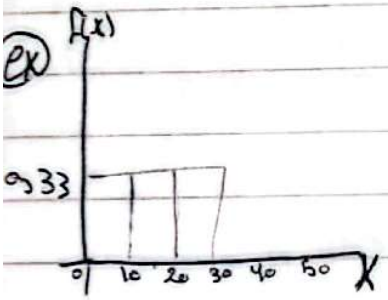


* $f(x) = \frac{1}{b-a}, a \le x \le b$

* $\mu = \frac{a+b}{2}$

* $var(\sigma^2) = \frac{(b-a)^2}{12}$

ex

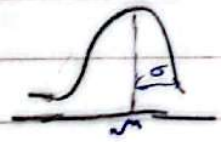


$\sigma = \sqrt{\frac{(30-0)^2}{12}}$

$\mu = \frac{a+b}{2} = \frac{0+30}{2} = 15$

$P(10 \le X \le 20) = (10)(\frac{1}{30-0})$

⟹ Normal dist.



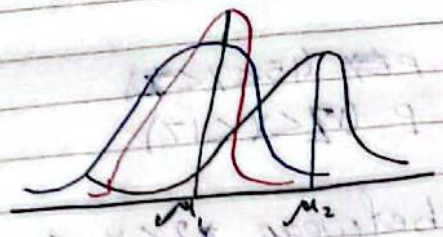
if $X \sim N(\mu, \sigma^2)$

$$\boxed{f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}}$$

$-\infty < x \infty$

$\pi = \frac{22}{7} \approx 3.141159$

$e \approx 2.1718,718$



* $X-\mu$ → Centered value
* $\frac{X}{q}$ → Scaling value of X
* $\frac{X-\mu}{\sigma}$ → standardized value of X

$$p(2<X<5) = \int_2^5 f(X)\,dX$$

$$Z = \frac{X-\mu}{\sigma}$$

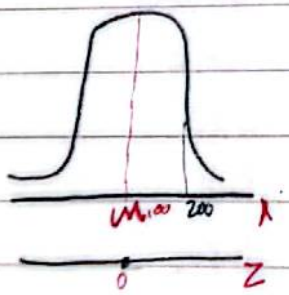
$E(X-\mu)$, $var(X) = \sigma^2$

$$E(Z) = \frac{E(X-\mu)}{\sigma} = \frac{E(X)-\mu}{\sigma} = \frac{\mu-\mu}{\sigma} = 0$$

$Var(Z) = 1$

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

ex



$p(100 < X < 200)$